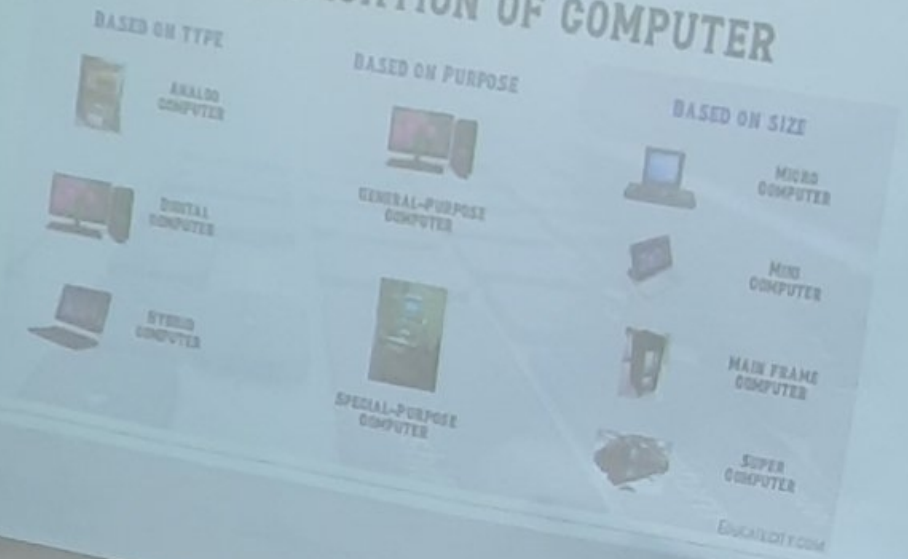
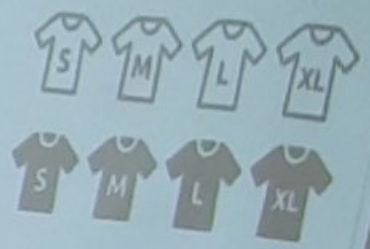
Taxonomy of Computers

- Computers are classified based on
  - According to Size
  - According to Technology
  - According to Purpose





11:23 AM Fri 17 Jan

All tools Edit Export E-Sign

Computer Classification

by

Size

7

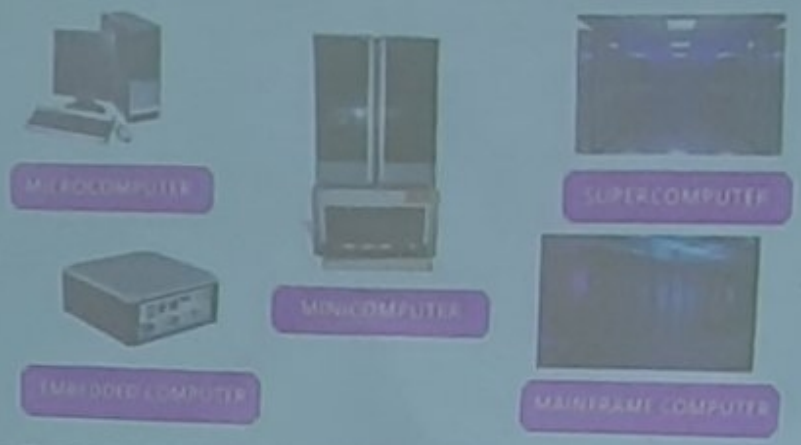
35

11:24 AM Fri 17 Jan

All tools Edit Export E-Sign

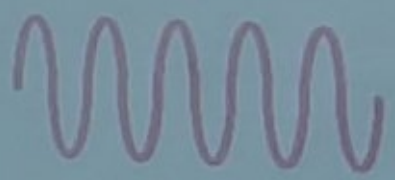
Classification of Computers According to size

- Mainframe Computers
- Workstations
- Minicomputers
- Microcomputers, or Personal Computers
- Supercomputers



Analog Computers

- *An analog computer or analogue computer is a type of computer that uses the continuous variation aspect of physical phenomena such as electrical, mechanical, or hydraulic quantities (analog signals) to model the problem being solved.*
- It processes analog data, stores data in a continuous form of physical quantities, and performs calculations with the help of measures.
- *Analog computers always take input in form of signals.*
- The input data is not a number in fact a physical quantity like temperature, pressure, speed, velocity.



11:27 AM Fri 17 Jan
All tools Edit Export E-Sign

Analog Computers

- Analog computers were especially well suited to simulating dynamic systems; such simulations could be conducted in real time or at greatly accelerated rates, thereby allowing experimentation by repeated runs with altered variables.
- They were widely used in simulations of aircraft, nuclear power plants, and industrial chemical processes.
- Other major uses included analysis of hydraulic networks (e.g., flow of liquids through a sewer system) and electronics networks (e.g., performance of long-distance circuits).
- By the 1970s, analog computers had been replaced by faster, more powerful digital computers.

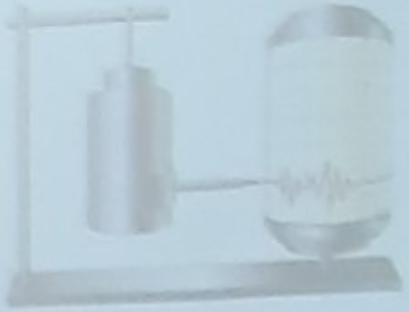
11:29 AM Fri 17 Jan

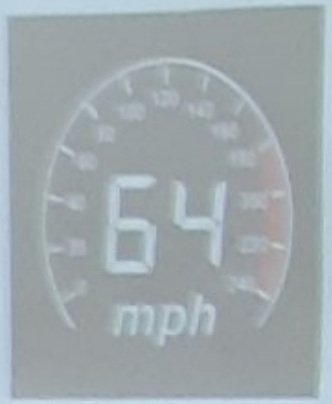
All tools Edit Export E-Sign

Example of Analog Computers

- Analogue Clock
- Automobile speedometer
- Thermometer
- Seismometer
- Voltmeter
- Operational amplifiers
- Mechanical integrators
- Slide rules
- Electric integrators, etc.

Examples of Analog Computer





Features of an Analog Computer

- Analog computer measures and answer the questions by the method of **"HOW MUCH".**
- Signals are continuous of (0 to 10 V)
- Accuracy 1% Approximately
- High speed
- Output is continuous
- Time is wasted in transmission time

Advantages and Disadvantages of Analog Computer

Advantages

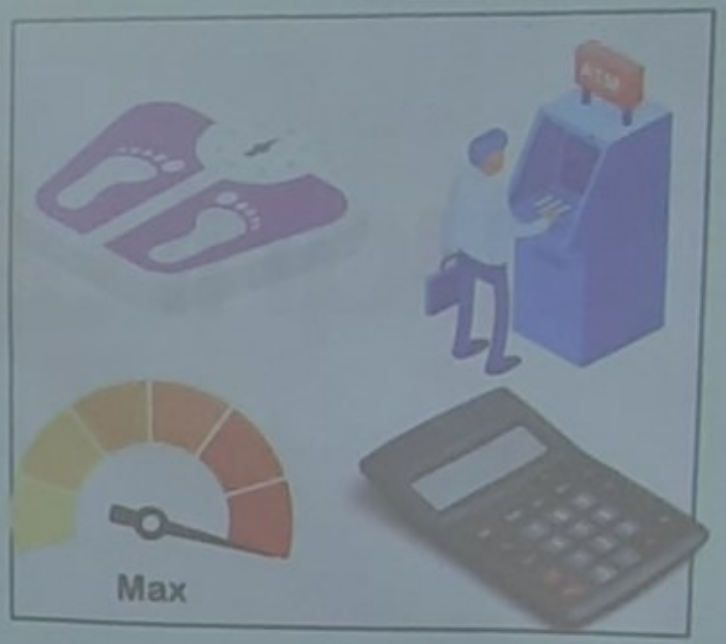
- Real-time Processing
- Efficient for Specific tasks
- Intuitive Representation
- Less prone to data errors

Disadvantages

- Limited Accuracy
- Lack of Versatility
- Complex Maintenance
- Susceptibility to Environment

Digital Computer

A digital computer is an electronic device that uses binary code to perform a wide range of tasks, from simple calculations to storing, retrieving, and processing data.

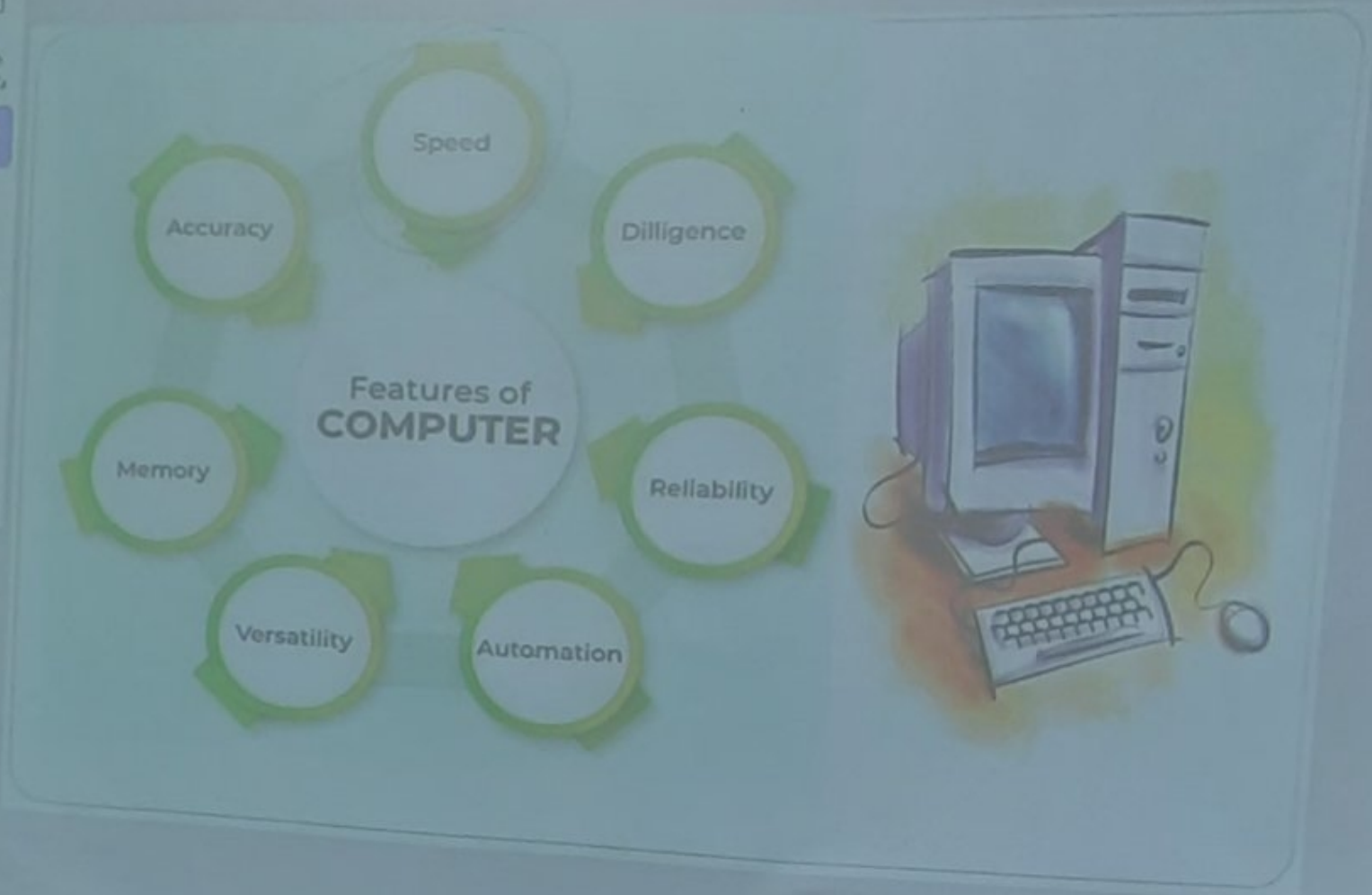


Digital Computers

- Any class of a digital devices capable of solving problems by processing information in discrete form.
- Digital computer counts and answer the questions by the method of "HOW Many".
- The input data is represented by a number.
- These are used for the logical and arithmetic operations.
- Signals are two level of (0V or 5 V)
- Accuracy unlimited
- *Output is continuous but obtain when computation is completed.*

Digital Computer

A digital computer is an electronic device that uses binary code to perform a wide range of tasks, from simple calculations to storing, retrieving, and processing data.


Features of
COMPUTER
Speed
Accuracy
Dilligence
Memory
Reliability
Versatility
Automation

11:44 AM Fri 17 Jan
All tools Edit Export E-Sign
96%

Example of Digital Computers

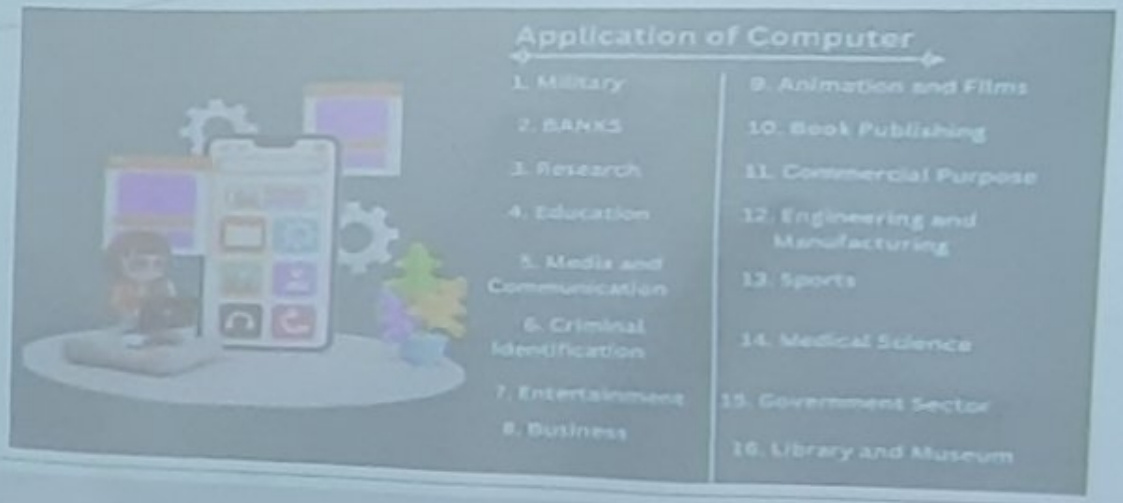
- Desktop
- Workstation
- Notebook
- Tablet PC
- Handheld computer (PDA)
- Smart Phone

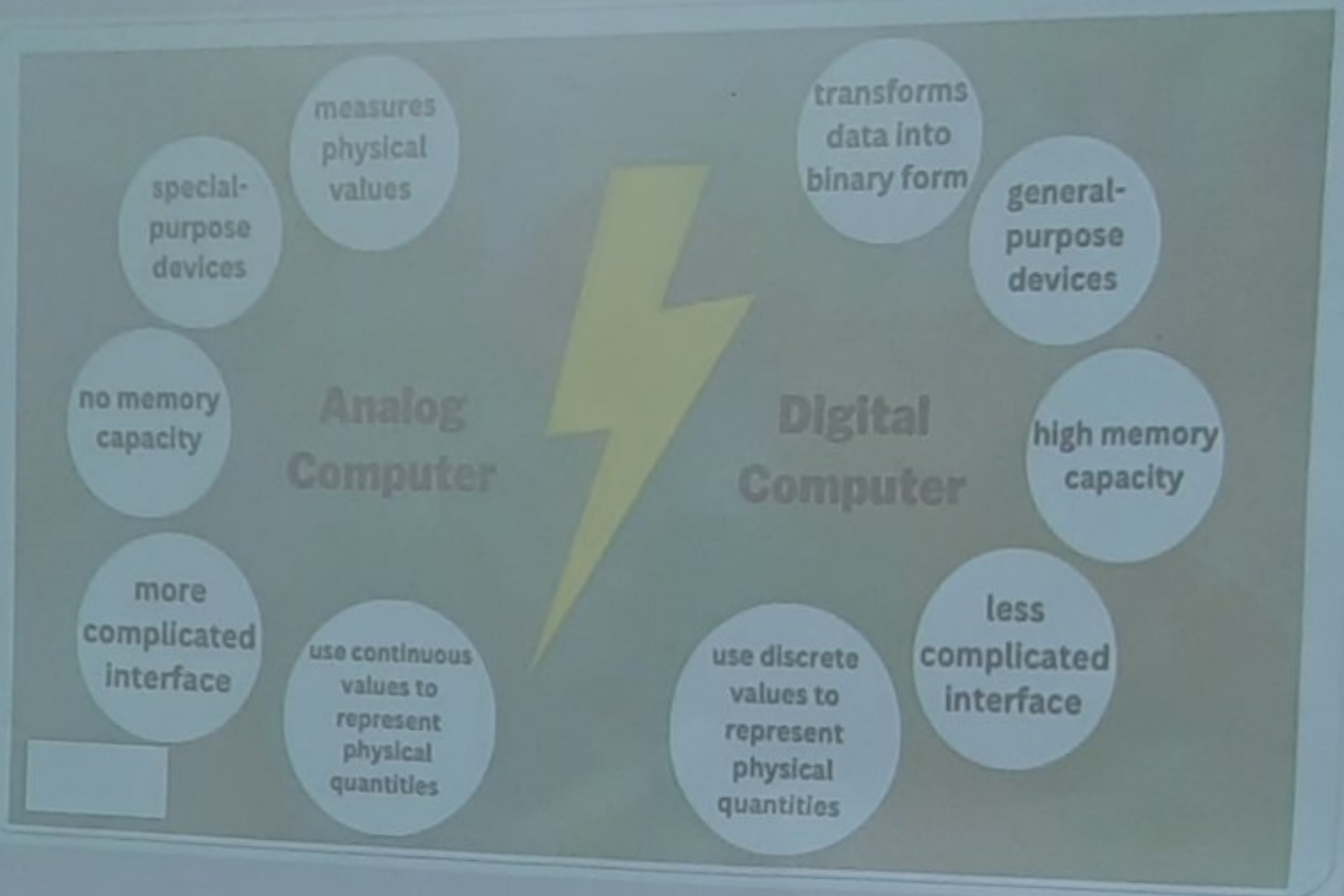
11:46 AM Fri 17 Jan
96%
All tools Edit Export E-Sign

Example of Digital Computers

Application of Digital Computers

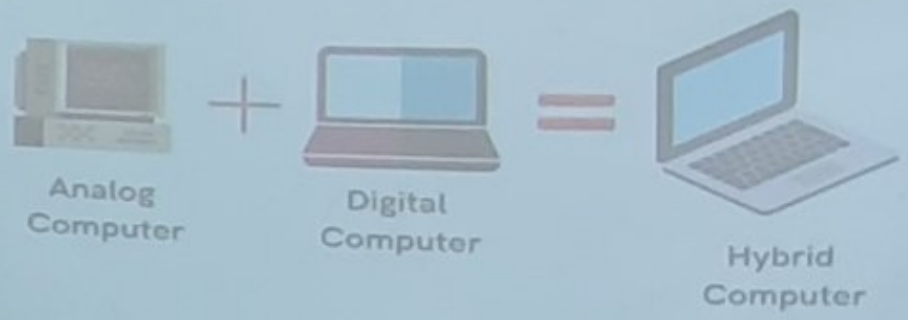
- Digital electronic computers for storing data, such as photos, music, documents, for performing complex mathematical computations, or for communication, commonly over a worldwide computer network.
- *All modern applications.*




Analog Computer
special-purpose devices
measures physical values
no memory capacity
more complicated interface
use continuous values to represent physical quantities
Digital Computer
transforms data into binary form
general-purpose devices
high memory capacity
less complicated interface
use discrete values to represent physical quantities

Hybrid Computers

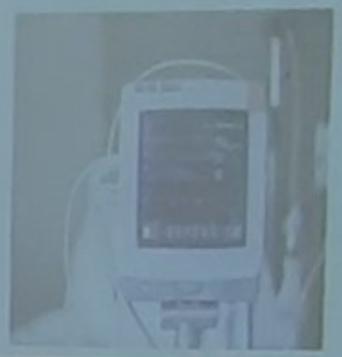
- Hybrid computers are computers that exhibit features of analog computers and digital computers.
- The digital component normally serves as the controller and provides logical and numerical operations,
- while the analog component often serves as a solver of differential equations and other mathematically complex problems.

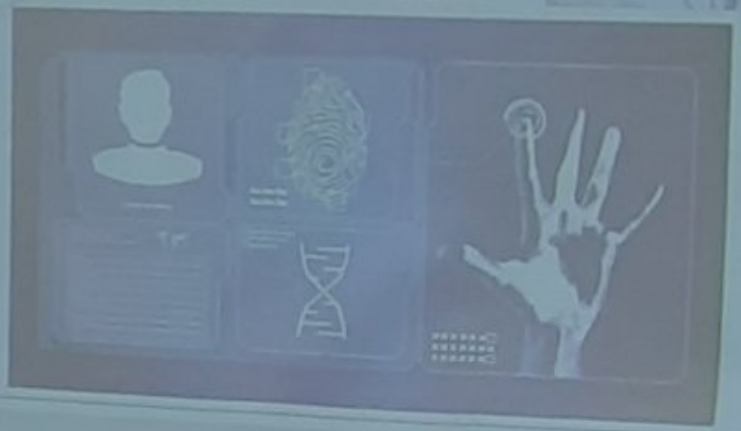


Use of Hybrid Computers

- Hybrid computers are used in hospitals to measure the heartbeat of the patient.
  - Electrocardiogram Machine
  - Ultrasound Machine
  - Monitoring Machine
  - Forensic Investigation Device, etc.

Examples of Hybrid Computers





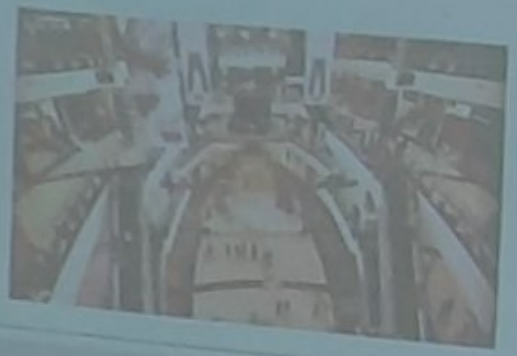
Computer Classification

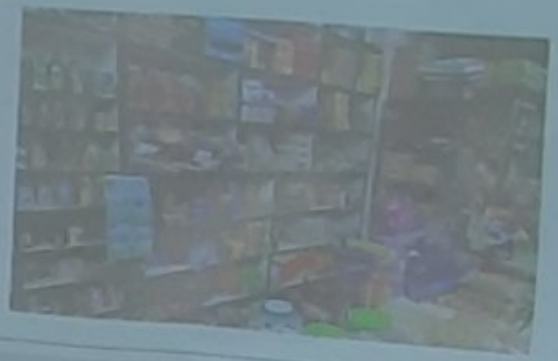
by

Purpose

Classification of Computers According to Purpose

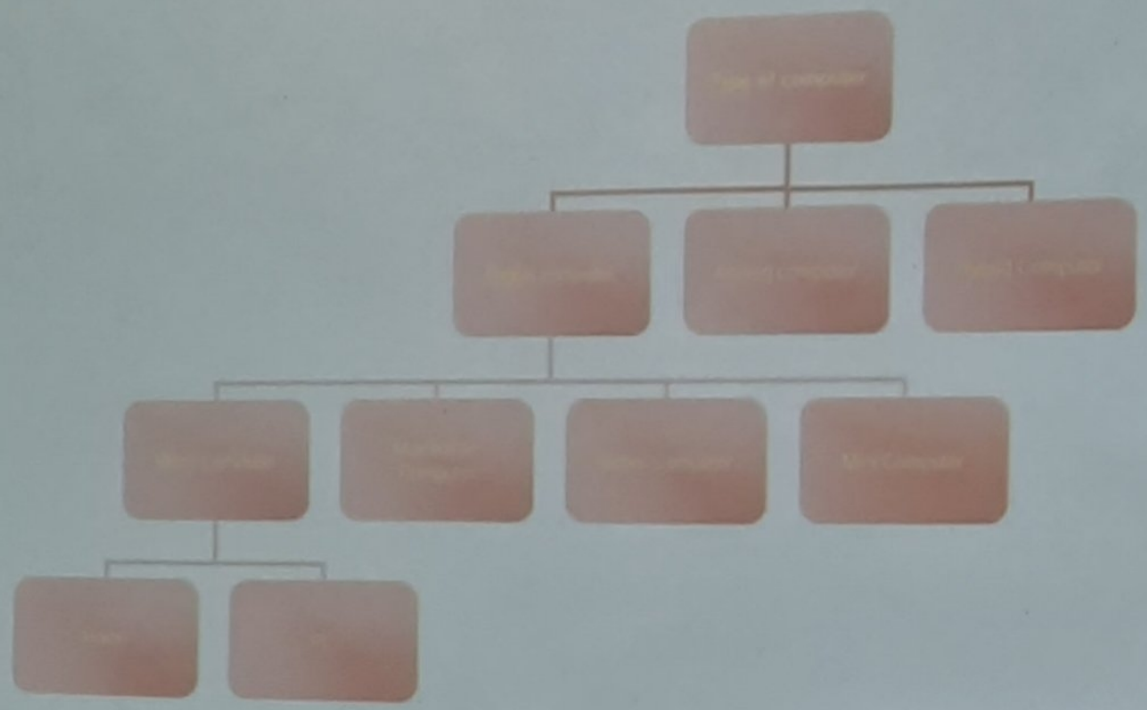
- General Purpose Computers
- Special Purpose Computers





Special Purpose Computers

- A computer that is designed to operate on a restricted class of problems.
- Use special purpose computer equipment to
- Obtain patient diagnostic information-ECG machines
- CCTV
- ATM, etc.

Fri 17 Jan
All tools Edit Export E-Sign

Classification of Computers: Summary